Inequalities: Polynomials

Questions 1–23

Question 1.

a) Sketch the graph of

$$y = (x-2)(x+1)(x+6)$$

b) Find the values of x, in set notation, where:
   i) $y > 0$
   ii) $y \leq 0$

c) Find the values of x, in interval notation, where:
   i) $y < 0$
   ii) $y \geq 0$

Question 2. Solve the following inequalities, writing your answers in set notation:

a) $(x+1)(x+2)(x+4) > 0$
b) $(x-2)(x-3)(x+5) < 0$
c) $(2x+1)(x+4)(x-8) \geq 0$
d) $(3x-2)(2x-5)(x+2) \leq 0$
e) $(2x-1)(x^2+9x+20) > 0$
f) $(2x-3)^2(x+1) > 0$
g) $(5x+2)x^2 \leq 0$

Question 3. Solve the following inequalities, writing your answers in set notation:

a) $(1-x)(x+5)(3+x) > 0$
b) $(x-7)(5-x)(x+1) < 0$
c) $(3x+4)(x-2)(1-x) \geq 0$
d) $(2-3x)(2x+1)(4-5x) \leq 0$
e) $(3-2x)(x^2-x-2) < 0$
f) $(4-5x)(3-x)^2 > 0$
g) $2x^2(4-7x) \leq 0$

Question 4. Solve the following inequalities, writing your answers in interval notation:

a) $x^3+5x^2-18x-72 > 0$
b) $4x^3-8x^2+3x \leq 0$
c) $9x^3-3x^2-5x-1 \geq 0$
d) $4+28x-9x^2-63x^3 < 0$
e) $6x^2-5x^3 \geq 0$
f) $8+61x+104x^2-48x^3 < 0$

Question 5. Solve the following inequalities, writing your answers in set notation:

a) $(1+x)(3-x)(x+5) \geq (1+x)(x-3)$

b) $(2+x)(x-4)x \leq (4-x)x^2$

c) $(5-x)(3+x) > (3+x)^2(x-5)$

d) $(4-3x)^2(6-5x) < (3x-4)(5x-6)^2$

Question 6. Given that $k > 3$, solve the following inequalities, writing your answers in set notation:

a) $(x-1)(x+1)(x-k) > 0$

b) $(x+3)(2x+5)(2x-k) < 0$

c) $(x+k)(2x+k)(x-2k) \geq 0$

d) $(kx+4)(kx-3)(3x+5) \leq 0$

Question 7. It is given that the inequality

$$(6x^2 + 19x + p)(x + 2) \geq 0$$

has the solution

$$\left\{x : x \in \mathbb{R}, -\frac{7}{2} \leq x \leq -2\right\} \cup \{x : x \in \mathbb{R}, x \geq q\}$$

where p and q are integers.
Find the values of p and q

Question 8. It is given that the inequality

$$px^3 + 17x^2 + qx - 72 \leq 0$$

has the solution $x \in (-\infty, -4] \cup \left[-\frac{9}{4}, r\right]$ where p, q and r are integers.
Find the values of p, q and r

Question 9. It is given that $f(x) = (3 + 2x)(2 + x)^2$

a) Solve $f(x) > 0$, writing your answer in interval notation.

$y = f(x)$ is mapped to $y = g(x)$ by a reflection in the x-axis.

b) Solve $g(x) > 0$, writing your answer in interval notation.

Quartic Inequalities

Question 10. Solve the following inequalities, writing your answers in set notation:

a) $(x - 1)(x - 2)(x + 3)(x + 5) > 0$

b) $(x + 4)(x + 3)(x + 2)x \leq 0$

c) $(2x - 3)(3x + 5)(x^2 - 1) \geq 0$

d) $(x^2 - 4)(9x^2 - 1) < 0$

e) $(4x^2 + 8x - 5)(3x^2 - 10x + 3) > 0$

f) $(x - 6)(x - 7)(2x + 1)^2 \geq 0$

g) $(2x + 7)^2(3x + 8)^2 > 0$

h) $(3x - 2)^3(x + 3) < 0$

Question 11. Solve the following inequalities, writing your answers in set notation:

a) $(2-x)(x+1)(x-3)(x+5) > 0$
b) $(4+x)(5-x)(x+3)(x-2) \leq 0$
c) $(4x^2-1)(1-9x^2) < 0$
d) $(1+x-2x^2)(9x^2-6x-8) \leq 0$
e) $(3x+1)^2(2-x)(2x-1) > 0$
f) $(3-x)^2(4-3x)^2 > 0$
g) $(7x+3)^3(4x+5) \geq 0$

Question 12. Solve the following inequalities, writing your answers in interval notation:

a) $x^4-5x^3-13x^2+53x+60 \geq 0$
b) $4x^4+28x^3+51x^2+27x < 0$
c) $259x^2+32x-48-2x^3-16x^4 \leq 0$
d) $15x^4+13x^3-6x^2 \geq 0$
e) $30x^2+325x+375-36x^3-8x^4 < 0$

Question 13. Solve the following inequalities, writing your answers in set notation:

a) $(x+2)(x-3)(x+1)(x-5) \geq (x+2)^2(x-3)(x+1)$

b) $(5+x)(x+6)(3-x)x \leq (x-3)^2x^2$

c) $(3-x)(2x+1)(4+x)^2 > (x-3)(2x+1)^2(4+x)$

d) $(7-2x)^3(4+9x) < (2x-7)^2(4+9x)$

Question 14. Given that $k > 3$, solve the following inequalities, writing your answers in set notation:

a) $(x-2)(x+3)(x+k)x > 0$

b) $(2x+1)(x-1)(3x-k)(2x+3) < 0$

c) $(x-k)(2x-k)(x-2k)(x+k) \geq 0$

d) $(kx-5)(kx+3)(x-2)(x+2) \leq 0$

Question 15. It is given that the inequality

$$(px^2 - 4x + q)(8x^2 + 6x - 9) < 0$$

has the solution

$$\left\{x : x \in \mathbb{R}, -\frac{3}{2} < x < -\frac{2}{5}\right\} \cup \left\{x : x \in \mathbb{R}, r < x < \frac{3}{4}\right\}$$

where p is an integer, and q and r are rational numbers. Find the values of p, q and r

Question 16. It is given that the inequality

$$px^4 - 12x^3 - 49x^2 + 87x + q < 0$$

has the solution $x \in \left(-\frac{5}{2}, -\frac{3}{2}\right) \cup (r, 4)$ where p, q and r are integers.
Find the values of p, q and r

Question 17. It is given that $f(x) = (5 - 3x)(2 + x)(x - 3)^2$

a) Solve $f(x) \leq 0$, writing your answer in interval notation.

$y = f(x)$ is mapped to $y = g(x)$ by a translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

b) Solve $g(x) \leq 0$, writing your answer in interval notation.

Question 18. It is given that $f(x) = (3x - 1)^3(2 - 5x)$

a) Solve $f(x) \geq 0$, writing your answer in interval notation.

$y = f(x)$ is mapped to $y = g(x)$ by a translation by the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$, followed by a reflection in the x-axis.

b) Solve $g(x) \geq 0$, writing your answer in interval notation.

Question 19. Solve the following inequalities, writing your answers in set notation:

a) $x^4 - 13x^2 + 36 > 0$

b) $4x^4 - 21x^2 - 25 \leq 0$

Hidden Polynomial Inequalities

Question 20. Solve the following inequalities, writing your answers in set notation:

a) $e^{3x} - 12e^{2x} + 41e^x - 30 \leq 0$

b) $e^{6x} - 6e^{4x} - e^{2x} + 30 > 0$

c) $e^{4x} - 4e^{3x} - 25e^{2x} + 88e^x - 60 \geq 0$

d) $4e^{8x} - 8e^{6x} - 59e^{4x} + 63e^{2x} + 180 < 0$

Question 21. Solve the inequality

$$2^{12x} - 10 \times 2^{9x} + 160 \times 2^{3x} - 256 \leq 0$$

writing your answer in interval notation.

Question 22. The inequality

$$(2\sin x - 1)(\sin x + 3)(\sin x + k) > 0$$

where $0 < x < 2\pi$ and k is a constant, has the solution $\left\{x : x \in \mathbb{R}, \frac{\pi}{6} < x < \frac{5\pi}{6}\right\}$
Find the possible values of k

Question 23. Solve the following inequalities, writing your answers in set notation:

a) $x^6 - 14x^4 + 49x^2 - 36 > 0$

b) $x^6 - 9x^3 + 8 \leq 0$

c) $x^6 - 9x^4 - 8x^3 + 72x > 0$